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RESEARCH ARTICLE

REVISED Reassessing the Education–FDI–Trade Nexus for a Sustainable Future: Evidence from Sub-Saharan Africa

[version 2; peer review: 1 approved with reservations]

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Abstract

Background

Education, foreign direct investment (FDI), and trade openness influence environmental sustainability through competing human capital, technology transfer, scale, composition, and pollution-haven channels. Evidence from Sub-Saharan Africa remains fragmented and often overlooks cross-sectional dependence and asymmetric responses.

Methods

This study examines a panel of Sub-Saharan African economies from 1990 to 2019, using production-based and consumption-based carbon dioxide emissions as environmental outcomes. The empirical strategy applies cross-sectional dependence and slope-heterogeneity diagnostics, second-generation unit root and cointegration tests, dynamic seemingly unrelated regression, continuously updated fully modified and bias-corrected estimators, asymmetric decompositions, and panel causality tests.

Results

Education is consistently associated with lower emissions across the principal long-run specifications. FDI is associated with lower production-based emissions but higher consumption-based emissions in the principal symmetric estimates, while the asymmetric findings show that the response differs between positive and negative FDI shocks and across time periods. Trade openness is predominantly emission-increasing, and urbanisation is associated with higher

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emission levels. Robustness analyses support the stability of core associations.

Conclusions

Sustainability policies in Sub-Saharan Africa should combine environmental education with screening and performance standards for FDI and trade-related production. Policies should consider asymmetric adjustments and economy-specific conditions when pursuing low-carbon development.

Keywords
education; foreign direct investment; trade openness; environmental sustainability; carbon emissions; Sub-Saharan Africa; panel data; asymmetry

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REVISED **Amendments from Version 1**

This revised version addresses the reviewer's methodological and interpretive concerns while retaining the study's original objectives, data, and empirical framework. The description of the sample has been clarified as a balanced panel of 50 Sub-Saharan African economies observed from 1990 to 2019, comprising 1,500 country-year observations. The education variable is now explicitly defined as government education expenditure—a fiscal-input proxy—rather than a direct measure of educational attainment, literacy, or human capital. The model specification has also been clarified to identify urbanisation and gross capital formation as control variables.

The interpretation of the empirical findings has been revised to correspond precisely with the coefficient signs reported in the tables, particularly the contrasting effects of foreign direct investment on production-based and consumption-based CO_2 emissions. The discussion of positive and negative shocks now distinguishes between short-run and long-run asymmetries, and the Wald-test results are interpreted systematically. Additional explanations address endogeneity, reverse causality, estimator choice, emissions-accounting differences, and country-level heterogeneity. The limitations and policy implications have been revised accordingly, and the funding information has been added.

Any further responses from the reviewers can be found at the end of the article

Introduction

The ecological status of the planet has never been more susceptible to sustainable management practices than it is at this moment in the history of modern society, facing an emergency of unparalleled scale. It is a common occurrence at all levels, and education can be a good channel to mitigate it (Kumar et al., 2023). The instability and value of the financial rapport between educational pursuits and environmental sustainability explain the need to establish a more sustainable future for the world. The crisis of CO_2 emissions into the atmosphere is one of the most significant problems of the twenty-first century that affects economic growth, health, and environmental sustainability on a global scale. The major factors of carbon dioxide emissions are anthropogenic, especially deforestation, industrial activities, and burning fossil fuels (Lone & Bhat). It is clear that emissions from the burning of fossil fuels are one of the primary sources of greenhouse gases (GHGs) and that the emission of CO_2 into the atmosphere contributes to global warming, storms that have increased in number, rising sea levels, and disrupted ecological stability, which are substantial contributors to climate change (Soeder, 2021). These changes are not limited to environmental impact but also influence the social and economic fields. Furthermore, because the climate system is interconnected, the carbon dioxide emissions in a specific area will inevitably have an effect in other regions; the CO_2 in the atmosphere affects all countries, regardless of the political boundaries. In turn, it is crucial to understand that the effects of domestic emissions go global, and such phenomena as carbon leakage also need to be considered (Pan et al., 2020). To start solving the environmental concern of excess carbon dioxide emissions, nations must examine the relationships between carbon emissions, foreign direct investment, trade openness, and education.

The complex factors underlying carbon dioxide emissions are closely intertwined with economic, ecological, and social processes. The constituent components that contribute to the overall carbon footprint do not make equal contributions. Among them, education (EDU) (Afolayan et al., 2020) and foreign direct investment (FDI) (Huang et al., 2022) are the most relevant. The other impact of emissions can be influenced by capital flows, such as FDI, which are closely related to increased economic activity. Through educational efforts, much has been done to advance society's trustworthiness and progress towards greater sustainability. There are also preconditions for the intensification of emissions due to industrial and commercial growth, trade openness (TO) (Jun et al., 2020), and economic growth (EG) (Wang et al., 2022). Their energy mix significantly influences countries' emission profiles; countries with resources tend to use fossil fuel. Urbanisation (UR) can boost economic growth but can also increase energy use (Ali et al., 2020). The ecological footprint (EF) (Ali et al., 2020) and technological development (TD) (Xu and Lan, 2023) are also represented as emission patterns. The regulations (RQ) and institutional structures (IP) of a specific country determine its ability to develop and adopt policies that enhance environmental conditions (Huang et al., 2022; Ali et al., 2020). Although the effect of outwards foreign direct investment (OFDI) on a host country is disputed (some researchers may find potential advantages in the form of economic growth and stabilisation in the case of influence on main spheres, Zafar et al., 2022), its influence on emissions remains unclear. Therefore, the current analysis asks whether it (EDU) is the mechanism for environmental sustainability and whether FDI and trade openness (TO) are. Investing in education, FDI, and trade liberalisation are interconnected and aim to overcome the difficulties faced by scholars and policymakers (Aust et al., 2020). Recent literature shows that investment in education enhances environmental safety, and FDI and openness to trade can contribute to environmental degradation (Tiba and Belaid, 2020). Providing more evidence, education (EDU) has become a determinant, as it has an influential effect on society and the economy.

Education (EDU) has become a determining variable with a substantial impact on the economy and society. Afolayan et al. (2020) and Xu and Lan (2023) examined the relationship between education and CO_2 emissions. Education can

strengthen human capital and support informed, environmentally responsible choices; however, education-led income and production growth may increase energy demand. Zafar et al. (2022) emphasised the role of education in adopting modern energy-efficient technologies. FDI can also increase emissions through scale and pollution haven effects or reduce them through cleaner technology and management practices (To et al., 2019; Hanh et al., 2020; Hao et al., 2021). Trade openness can stimulate industrialisation and transport activity while facilitating access to cleaner inputs and technologies (Wang & Zhang, 2021; Jun et al., 2020).

However, free trade facilitates the transition to sustainable finance and demonstrates environmentally friendly technologies. Wang and Zhang (2021) recognised that trade policies can mitigate the effects of emissions because they influence traded goods. The looming environmental issues in Sub-Saharan Africa (SSA) have heightened the urgency of this research. Economic growth and environmental protection must be delicately balanced because the region is focused on economic growth. The main predictors of this balance are education, foreign direct investment (FDI), and openness to trade. The interaction between these factors is the focus of the analysis, which is presented in detail to enhance policy quality. This study is anticipated to fill gaps in the literature by critically analysing the relations using advanced econometric tools and thus provide subtle and precise information.

Although empirical studies on environmental sustainability are increasingly numerous, research findings on the roles of education, foreign direct investment (FDI), and openness to trade remain dispersed and fragmented. The literature documents the emission-cutting and emission-enhancing effects of these organisations, and this lack of certainty hinders policymakers from developing consistent policies that can balance development and environmental conservation, particularly in developing nations. Educationists and environmentalists who have researched the convergence between education and environmental sustainability have arrived at various conclusions. Some researchers have suggested that more education results in increased consumption and activity among people, thereby contributing to greater carbon emissions. According to other studies, education enhances environmental awareness, the adoption of cleaner technologies, and the overall level of public support for environmental protection, leading to decreased pollution. These are mutually conflicting arguments that are rarely addressed in a single empirical model, and most studies assess education alone without considering other macroeconomic variables. The same discrepancies prevail in the foreign direct investment (FDI) literature. The pollution haven hypothesis assumes that FDI leads to increased emissions because more polluting industries are transferred to countries with weaker environmental regulations. The pollution halo hypothesis, on the other hand, tends to focus on the movement of technology, efficiency gains, and the spread of superior management practices, leading to lower pollution. Each of the two positions is supported by empirical data, often based on sample selection, estimation techniques, or control variables. However, it is not the subject of most research to determine the interactions among the environmental impact of FDI, human capital, and trade exposure.

The results for trade openness are also conflicting. The scale effect presupposes greater emissions because increased production and transportation are expected. The effects of composition and technique imply that openness may lower emissions by shifting production towards cleaner industries and spreading advanced technologies. This is because, in most empirical literature, the mechanisms are tested individually and based on average effects, which conceal the possibility of asymmetries and context-dependent effects.

The main literature gap is the lack of an integrated analytical approach that simultaneously examines education, FDI, and trade openness and addresses issues of heterogeneity, asymmetry, and cross-sectional dependence. These mostly used single-equation models or first-generation panel estimators, resulting in studies that captured complex interdependent effects within countries and identified both positive and negative effects. Subsequently, conflicting results persist, and policy interpretations are poor.

This study aims to fill the gap in the literature by combining education-based human capital, foreign direct investment, and trade openness for environmental sustainability. The framework explicitly leaves room for asymmetric responses and second-generation panel methods for long-run dynamics analysis. Thus, this study balances theoretical debates and provides additional evidence on the circumstances in which education, FDI, and trade can contribute to environmental enrichment or degradation. This strategy is especially applicable to sub-Saharan Africa, where rapid economic integration and human capital development are occurring alongside rising environmental pressures.

Literature review and hypothesis development

Theoretical foundation

Environmental sustainability is a discourse that is becoming increasingly vital and will continue to gain momentum in the future. It expresses the urgency of being efficient in marshalling resources while simultaneously protecting the natural ecosystems on which humanity depends. In its conventional sense, environmental sustainability (ES) refers to efforts to

ensure that the current generation can fulfil its needs today and that future generations will have the capacity to fulfil their needs. This knowledge is grounded in the broader concept of sustainable development, which requires the incorporation of environmental stewardship, social equity, and economic development. Natural resources and ecosystems play a central role in society's welfare, and their assessment is directly linked to the community's overall well-being (Ziaul & Wang, 2023).

Empirical studies provide conceptually consistent explanations of the determinants of ES and are deeply rooted in economic theory. Human Capital Theory teaches us that the more knowledge a person is provided with, the better prepared they are to handle the environment. A highly educated population is thus better acquainted with sustainable practices and is willing to adopt new environmental protection strategies positively (Mourad, Wahid, Alkubise, and Najar, 2023). The neoclassical EKC theory argues that economic growth does not inevitably lead to environmental degradation, as long as enlightened citizens steer countries toward cleaner technologies and a more sustainable path of development. Studies conducted by Duodu et al. show how education can influence attitudes and behaviour to facilitate civic participation in environmental campaigns, as explained in terms of behavioural economics (Ekins, 2011).

Environmental sustainability is affected both positively and negatively by trade openness. The scale effect means that as industrial production and trade increase, pollution and resource consumption rise (Varghese, 2024). On the other hand, a change in the composition of the environment can be observed as a shift to cleaner industries or services producing what has been called the environmental composition effect. Rehman, Bhatti, Kraus, and Ferreira (2021) contend that stringent environmental standards might complement trade, and as a result, a race to the top may mobilise, in which high-income, environmentally neutral goods are swapped. There is an elegant correlation between ES and foreign direct investment (FDI). Duodu, Kwarteng, Oteng-Abayie, and Frimpong (2021) propose the Pollution Haven Theory, which suggests that multinational companies will set up in jurisdictions with weak environmental policies, thereby increasing pollution in the area. Conversely, the Pollution Halo Hypothesis states that FDI may also be beneficial to developing states by introducing new technology and improving environmental management systems. Governance plays a critical role, since host-country regulations are the main drivers of the net effect of FDI on sustainability.

Urbanisation presents both opportunities and challenges. Rapid urban development increases per capita pollution and resource exploitation, particularly in poorly planned cities (Bordun, Antonyak, Dumych, Hrytsyshyn, and Bilous, 2023). However, cities have economies of scale that help enhance social services and minimise waste in infrastructure (Bowles, Boetto, Jones, and McKinnon, 2018). City centres of higher learning and innovation encourage sustainability activities and nurture favourable laws and cultures. However, these objectives should be achieved through sustainable urban planning and policy frameworks (Ang, Choolani, and Poh, 2024). The four factors connected to education, trade, openness, and urbanisation interact to form complicated systems that impact environmental sustainability. These dynamics are important for identifying viable ways to support the provision of ecosystem services. Future governance should adopt technologies that can realise these strategies.

- Education and environmental sustainability

Education is a catalyst for growth and prosperity, as it is an investment in individuals and society in general. It has long been known to be an important factor in human development and economic growth. In recent years, scholars have received growing interest in two-way communication between the environment and education. Further elaboration of this relationship shows that it is a prerequisite for sustainability in practice, technological innovation, policy advocacy, diversifying the economy, and exchanging best practices globally. The procedure highlights the immense ability of education to influence cognition and create a globally conscious society that is sustainably aware of everyone. However, two empirical studies present opposing results: the first shows that education has a positive relationship with emissions, and the other indicates the opposite.

The quantitative data for the first group illustrate that education and CO_2 emissions are positively correlated. Afolayan et al. (2020) concluded that a high level of education allows the economy to be active, which in turn pushes the size of aggregate emissions. Similarly, Xu and Lan (2023) found that both education and economic growth positively impact CO_2 emissions. Continuous air pollution may be explained by the further use of fossil fuels. Suman (2021) suggested that the prevalence rates of greenhouse gases depend on the level of education and job opportunities at the regional level. Moreover, Zafar et al. (2022) hypothesised that education is also one of the major causes of environmental degradation. The latter group highlights the adverse relationship between education and CO_2 emissions. Leal Filho et al. (2019) noted that energy sustainability is a top-priority agenda among universities in an effort to limit carbon emissions and lower energy use in buildings. These sustainability programs require huge investments by many institutions of higher learning. As illustrated by Yin, Jiang, Lin, and Liu (2022), online education has a twofold effect on carbon emissions: online classes

reduce commuting-related emissions because most of them can be taken at home; however, online education leads to emissions through the use of devices and servers. Chen et al. (2022) argued that education enables personal adaptation to climate change and the creation of technologies that reduce greenhouse gas emissions. According to Zafar et al. (2020), education positively affects carbon emission reduction.

- Foreign direct investment and environmental sustainability

Two different dichotomies have been developed in the literature with respect to the nexus between foreign direct investment (FDI) and greenhouse gas emissions. An extensive body of literature records a favourable relationship, that is, increased FDI inflows are correlated with emissions of carbon dioxide. According to Yuan Wang and Huang (2022), a positive initial shock in short-term increases in FDI can trigger an increase in CO_2. Expanding on the pollution haven hypothesis, Huang et al. (2022) argue that FDI is more concentrated in jurisdictions where environmental regulations are relatively lenient, thus adding up to 525 percentage points of emissions by 2100, depending on geographic and institutional factors. According to Hao et al. (2021), this effect is explained by the emissions produced during different levels of production and consumption. It is important to mention that the impact of the environment is considerably softened by the investor sphere, technology, and motivation. Huang et al. (2022) emphasise the role of macroeconomic variables, namely economic development, financial development, urbanisation, and tourism, in mediating the indirect impact of FDI on carbon emissions. The (quantitative) relationship was measured by Wang et al. (2022), and the coefficient estimates the result of a 1% increment of FDI resulting in a 0.03% increment in emissions. However, the error correction coefficient has a negative value, which means that, in the short term, the system aims to return the equilibrium situation to its original condition.

However, a substantial body of research suggests that FDI may lead to technological spillovers that reduce carbon emissions. To et al. (2019) showed that FDI contributes to the spread of clean technology innovations, thereby helping prevent emissions. They argue that host countries with higher education levels disproportionately benefit from the environmental returns of such inflows. Q. The same argument was supported by Wang et al. (2023), who reported that in better-quality countries, there is a higher likelihood of a reduction in the level of CO_2 emissions that can be linked to FDI. Hanh et al. (2020) found that FDI tends to reduce pollution, primarily by providing environmentally friendly technologies that decrease energy consumption and carbon emission intensities. As stated by Apergis, Pinar, and Unlu (2023) and Ren, Yuan, Ma, and Chen (2014), the potential of FDI to promote the proliferation of clean technology is decisive in mitigating greenhouse gas emissions; thus, policy frameworks aimed at capitalising on positive externalities are critical.

- Trade openness and environmental sustainability

Jun et al. (2020) postulate that high levels of trade openness may strengthen the dominance of specific industries, thereby leading to higher pollution. Chhabra, Giri, and Kumar (2023) revealed that the production of goods is an indicator of carbon emissions. Additionally, the exchange of products and services can significantly impact the adoption of environmentally friendly technologies, helping countries transition to more sustainable economic activities. Nonetheless, when goods are produced using contaminated products, environmental degradation occurs. There is even a possibility that trade openness can reduce carbon emissions in wealthier nations, as described by Wang and Zhang (2021).

Furthermore, Ali et al. (2020) found that liberal trade policies can predispose an environment to degradation because nations may loosen their regulations in the pursuit of trade and investment. Liberalisation forces energy-intensive industries to relocate to places with less stringent environmental regulations, thereby increasing carbon emissions. Two distinct claims support this relocation trend, suggesting a common underlying mechanism. Another academic research stream has introduced a negative dependent correlation between trade openness and carbon dioxide levels. Several reports argue that liberalised trade may reduce carbon emissions in developed nations. For example, Karedla, Mishra, and Patel (2021) found that CO_2 emissions in India decreased significantly as the country opened up to international trade. Accordingly, Lin and Umetsu noted that carbon dioxide emissions are expected to decline in newly industrialised countries as they become more open to trade. According to Chhabra, Giri, and Kumar (2022), increased trade facilitates the exchange of knowledge and technology, thereby improving environmental performance by reducing carbon dioxide emissions. Finally, Thi, Tran, and Nguyen (2023) argued that emissions can be reduced by utilising renewable energy sources, increasing trade openness, and fostering innovation. A summary of the literature is provided in Table 1.

Figure 1 presents the theoretical foundation of this study, and Table 1 summarises the literature survey.

Table 1. Summary of the literature survey.

Author	Sample (year)	Methodology	IV(s)	EDU	FDI	TO
Q. Wang, Yang, Li, and Wang (2023)	67 countries (1990–2019)	FMOLS			-VE	
Thi, Tran, and Nguyen (2023)	53 nations (19902–2019)	FMOLS, DOLS, GMM System estimator	REC, TO, INNO			-VE
Chhabra, Giri, and Kumar (2023)	BRICS (1991–2019)	DCCE	IQ, PS, PE			+VE
Apergis, Pinar, and Unlu (2023)	BRICS (1993–2012)	GMM	-		-VE	
Xu and Lan (2023)	China (2010–2021	DEA	TD	+VE		
Y. Wang and Huang (2022)	East Asia (2011–2020)	ARDL	TR, EG		+VE	
Zafar, Saleem, Destek, and Caglar (2022)	22 top remittance-receiving countries (1986–2017)	second-generation unit root techniques	remittances, export diversification	+VE		
Chen et al. (2022)	Global countries	Econometric models	-	-VE		
Chhabra, Giri, and Kumar (2022)	23 middle-income countries (1994–2018)	GMM	Inn			-VE
Hanh et al. (2020)	Vietnam and Indonesia (1960–2018)	ARDL	-		+VE	
Yin, Jiang, Lin, and Liu (2022)	China (2011–2019)	Regression model	-	-VE		
Huang et al. (2022)	G20 economies (1996–2018)	FGLS	ED, FD, UR, and RQ	+VE	+VE	
Hao, Ba, Ren, and Wu (2021)	China (2003–2016)	SYS-GMM	TR, OFDI		+VE	
Q. Wang and Zhang (2021)	182 countries (1990–2015)	Econometric models	-			-VE
Lin and Umetsu	NIC countries (1971–2020)	Econometric models	-			-VE
L. Wang et al. (2022)	NICs, (1990–2016)	ARDL-PMG	-		+VE	
Suman (2021)	Nepal	Econometric models	-	+VE		
Zafar, Shahbaz, Sinha, Sengupta, and Qin (2020)	OECD (1990–2015)	Second generation methodologies	NRA, FDI, EG	-VE		
Jun, Mahmood, and Zakaria (2020)	China	wavelet-coherence analysis	-			+VE
Ali, Yusop, Kaliappan, and Chin (2020)	OIC countries	DCCE, MG, PMG	TO, UR, EF, IP			+VE
Afolayan, Okodua, Oaikhenan, and Matthew (2020)	Nigeria (1980–2017)	ARDL	ED	+VE		
To, Ha, Nguyen, and Vo (2019)	Asian region (1980–2016)	DOLS, FMOLS	-		-VE	
Ren, Yuan, Ma, and Chen (2014)	China (2001–2011)	environmental regulation	EI, TO, CA, ER,		-VE	

Note: EDU = education; FDI = foreign direct investment; TO = trade openness. Signs summarise the direction of the reported relationship with environmental sustainability. Source: authors' compilation from the cited literature.

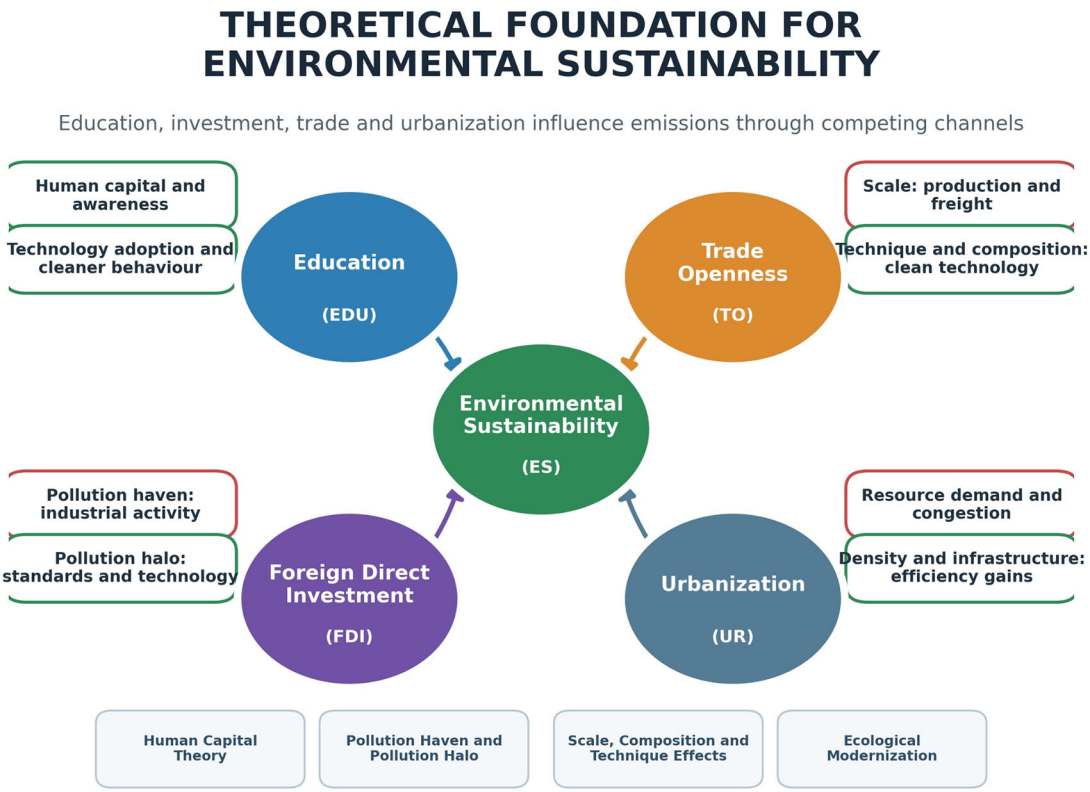


Figure 1. Theoretical foundation of the study. Education influences emissions through human capital and environmental awareness channels; foreign direct investment (FDI) operates through competing pollution-haven and pollution-halo mechanisms; trade operates through scale, composition, and technique effects; and urbanisation can amplify demand pressures or support efficiency. Arrows indicate hypothesised directions, with the net relationship determined empirically. Figures should be submitted separately as high-resolution TIFF or JPEG files (minimum 300 dpi) according to the journal template.

Methods

Study design, data and variables

This study uses annual panel data for Sub-Saharan African economies from 1990 to 2019. The reported estimation sample is a balanced panel of 50 cross-sectional country units observed for 30 years, yielding 1,500 country-year observations (Table 12). Units were retained when the variables required for the final specifications were available during the study period. The phrase 'top 50 polluted nations' in the earlier draft was inconsistent with the stated regional scope and has therefore been removed. The operational definitions, transformations, and original data sources for environmental outcomes, education expenditure, FDI, trade openness, urbanisation, and gross capital formation are reported in Table 2.

Data and methodology of the study

Model specification

This study assesses how the education expenditure proxy relates to environmental sustainability in a 50-country Sub-Saharan African panel over 1990–2019. The generalised equation is as follows:

$$ES|EDU \tag{1}$$

Equation (1) has been extended to include four macroeconomic indicators, following the existing literature on environmental degradation. The revised Equation (2) is as follows:

$$ES \perp EDU, TO, FDI, UR, GCF \tag{2}$$

Table 2. Proxy measures of the research variables.

	Notation	Definition	Data sources
Environmental degradation	ED	Consumption-based CO_2 emissions	WDI
		CO_2 per capita	
Education	EDU	Government education expenditure as a share of total government expenditure is a fiscal-input proxy for public commitment to education, not a direct measure of attainment, literacy, schooling years or realised human capital.	WDI
Trade openness	TO	Sum of imports and exports as % of GDP	UNCTAD
Foreign direct investment	FDI	Net inflows of FDI as a % of GDP	WDI
Urbanization	UR	Percentage of a country's population living in urban areas.	OurWorldinData
Financial deepening	FD	Ratio of private credit to GDP	OurWorldinData
Gross Capital Formation	GCF	The total value of investments (both public and private) in a country is often expressed as a percentage of the country's gross domestic product (GDP).	OurWorldinData

Note: CO_2 = carbon dioxide; C-CO_2 = consumption-based carbon dioxide; EDU = education; FDI = foreign direct investment; TO = trade openness; UR = urbanisation. Source: World Development Indicators, UNCTAD, and Our World in Data.

ES, EDU, TO, FDI, UR, and GCF represent environmental sustainability, education expenditure, trade openness, foreign direct investment, urbanisation, and gross capital formation, respectively. The logarithmic transformation of the variables in Equation (2) can be expressed as follows:

$$CO_2 = \beta 0 + \beta 1\,EDU + \beta 2\,TO + \beta 3\,FDI + \beta 4\,UR + \beta 5\,GCF + \varepsilon \quad (3)$$

where β0 is the intercept, β1, β2, β3, β4, and β5 are the coefficients associated with the explanatory variables, and ε is the error term.

The first coefficient (β1) captures the marginal association between the share of government expenditure allocated to education (EDU) and CO_2 emissions, holding other regressors constant. EDU is interpreted as a fiscal-input proxy for the priority assigned to education and the capacity to finance schools, teachers, curricula, and environmental learning. It is not equivalent to educational attainment, literacy, average years of schooling, education quality, or human capital stock. A negative coefficient is consistent with an indirect channel in which sustained public education expenditure supports environmental awareness and the capacity to adopt cleaner practices. However, the coefficient should not be interpreted as the effect of an additional year of schooling or as a direct change in individual knowledge.

The second coefficient in the equation, 02, represents the exact effect of a unit change in trade openness (TO) on future variations in CO_2 emissions, assuming that all other factors are constant. Variability may also be present in the sign of 2. Alternatively, with trade openness leading to increased international trade and transportation activity, a positive 2b, implying an increase in emissions, would be a reasonable outcome (Chhabra et al., 2023; Thi et al., 2023). However, β2 might be negative if the diffusion of cleaner technologies and the decrease in dependence on carbon-intensive production are enhanced by increased openness (Q. Wang & Zhang, 2021; Jun et al., 2020).

The third coefficient, 03, represents the extent to which foreign direct investment (FDI) is influenced by CO_2 emissions, holding all other factors constant. The 3(3) sign can vary according to the orientation of FDI. Coefficient 3 = − may be a negative value when FDI introduces ecologically friendly technology and sustainable methods of production (Wang and Huang, 2022; Hanh et al., 2020; Huang et al., 2022). However, when FDI invests in highly polluting industries, a positive correlation is expected (Hao et al., 2021; L. Wang et al., 2022).

The fourth coefficient (β4) captures the association between urbanisation (UR) and CO_2 emissions, holding the remaining variables constant. Urbanisation can increase emissions through housing, transport, construction, and energy demand; however, agglomeration and infrastructure efficiency can produce the opposite effect, reducing emissions. Gross capital formation (GCF) is included as an additional control because investment-driven expansion can change productive capacity, infrastructure, and energy demand independently of FDI. Therefore, the final controls capture two major

non-core channels: urban scale and density effects (UR) and domestic capital accumulation (GCF). Their inclusion follows the theoretical mechanisms developed in the literature and reduces the omitted variable bias, although it cannot eliminate it. The reported coefficients remain conditional associations because unobserved policy, sectoral, and energy-mix factors may still affect emissions.

The preferred specification retains EDU, FDI, and TO as the focal regressors, and UR and GCF as theoretically motivated controls. Financial deepening (FD), although defined during data screening, is not included in the final reported regressions; this distinction avoids treating every available indicator as part of the estimating equation.

Estimation strategies

To assess the empirical relationship between ED, EDU, TO, FDI, UR, and GCF, this study implemented several econometric tools, including the cross-sectional dependency test following Breusch and Pagan (1980), Pesaran (2004), Pesaran (2006), Pesaran, Ullah, and Yamagata (2008). The following equation is used to derive the test statistics:

$$y_{it} = \alpha_i + \beta_i x_{it} + u_{it} \quad \mathrm{i} = 1 \ldots \ldots \mathrm{N}, \mathrm{t} = 1 \ldots \ldots \mathrm{T} \tag{4}$$

$$LM = T \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \widehat{\rho}_{IJ_X^2 N(N+1)2} \tag{5}$$

The Lagrange multiplier (CD_{lm}) is the scaled version of the LM test in Eq (6):

$$CD_{lm} = \sqrt{\frac{N}{N(N-1)}} \sum_{I=1}^{N-1} \sum_{J=i+1}^{N} \left(T\widehat{\rho}_{ij} - 1 \right) \tag{6}$$

When N is larger than T, the CD_{lm} estimation is subject to size-related problems. Therefore, Pesaran (2006) proposed the following CD test: Eqs (7) and (8), which is suitable when N is greater than T:

$$CD = \sqrt{2T/(N(N-1)} \left(\sum_{i=1}^{N-1} \sum_{J=I+1}^{N} \widehat{P}_{ij} \right) \tag{7}$$

$$CD_{lm} = \sqrt{\frac{2}{N(N-1)}} \sum_{I=1}^{N-1} \sum_{J=i+1}^{N} \left(\frac{(T-K)\widehat{\rho}_{ij}^2 - u_{Tij}}{v_{Tij}^2} \right) \vec{d}(N,0) \tag{8}$$

The stationary properties of the research variables were assessed using the framework proposed by Pesaran (2007).

$$\Delta Y_{it} = \mu_i + \theta_i y_{i,t-1} + \gamma_i \overline{y}_{t-1} + \sum_{k=1}^{p} \gamma_{ik} \Delta y_{i,k-1} + \sum_{k=0}^{p} \gamma_{ik} \overline{\Delta y}_{i,k-0} + \tau_{it} \tag{9}$$

where $Y_{it} - 1$ and $\overline{y}_{t-1}$ are the lagged-level average and first-difference operators, respectively, and the CIPS unit root test is presented in Equation 11.

$$CIPS = N^{-1} \sum_{i-1}^{N} \partial_i(N,T) \tag{10}$$

where the parameter $\partial_i(N,T)$ explains the test statistics of $CADF$, which can be replaced as follows:

$$CIPS = N^{-1} \sum_{i-1}^{N} CADF \tag{11}$$

A cross-sectionally augmented version of the test statistics is presented.

$$CIPS = \frac{1}{N} \sum_{i=1}^{N} CADF \tag{12}$$

An error-correction-based panel cointegration test was performed to reveal long-run cointegration in the empirical equation, following Westerlund (2007).

$$\Delta Z_{it} = \partial_i' d_i + \varnothing_i \left(Z_{i,t-1} - \delta_i' W_{i,t-1} \right) + \sum_{r=1}^{p} \varnothing_{i,r} \Delta Z_{i,t-r} + \sum_{r=0}^{p} \gamma_{i,j} \Delta W_{i,t-r} + \epsilon_{i,t} \tag{13}$$

The group test statistics can be derived using Equations 14 and 15, respectively.

$$G_T = \frac{1}{N} \sum_{i-1}^{N} \frac{\varphi_i}{SE\varphi_i} \tag{14}$$

$$G_a = \frac{1}{N} \sum_{i-1}^{N} \frac{T\varphi_i}{\varphi_i(1)} \tag{15}$$

$$P_T = \frac{\varphi_i}{SE\varphi_i} \tag{16}$$

$$P_a = T\varphi_i \tag{17}$$

The equation for coefficient estimation by executing CUP-FM and CUP-BC is as follows:

$$\widehat{\beta_{cup}} = \left\lfloor \sum_{i=1}^{N} \left(\sum_{i-1}^{T} \widehat{y}_{it} + \left(\widehat{\beta}_{cup} \right) \times \left(x_{it} - \overline{X_i} \right) - T \left(\gamma_i' \widehat{\beta}_{cup} \right) \Delta f_{et} + \left(\widehat{\beta}_{cup} \right) \widehat{+ \Delta \mu ei} + \left(\widehat{\beta}_{cup} \right) \right) \right\rfloor \times \left\lfloor \sum_{i=1}^{N} \sum_{i=1}^{T} \left(x_{it} - \overline{X_i} \right) \left(x_{it} - \overline{X_i} \right) \right\rfloor^{-1} \tag{18}$$

The empirical analysis is extended by decomposing changes in EDU, TO, and FDI into cumulative positive and negative partial sums, while UR and GCF remain symmetrically controlled. Asymmetry is expected because an increase and an equivalent decrease do not necessarily operate through identical adjustment costs, policy responses, sectoral compositions, or technology channels. For example, new FDI may immediately expand pollution-intensive production, whereas an FDI contraction may reduce output before the capital stock adjusts. Similarly, trade expansion and contraction can alter production and embodied carbon flows differently. The asymmetric equation is as follows:

$$ES_t = \left(\pi^+ EDU_{1,t}^+ + \pi^- EDU_{1,t}^- \right) + \left(\beta^+ TO_{1,t}^+ + \beta^- TO_{1,t}^- \right) + \left(\gamma^+ FDI_{1,t}^+ + \gamma^- FDI_{I1,t}^- \right) + \varepsilon_t \tag{19}$$

where $\pi+$, $\pi-$, $\beta+$, $\beta-$, and $\gamma+$, $\gamma-$ represent the long-run asymmetric coefficients of education, trade openness, and foreign direct investment, respectively. The decomposition of the explanatory variables can be derived as follows:

$$\begin{cases} \mathrm{POS}(\mathrm{EDU})_{1,it} = \sum_{M=1}^{W} \mathrm{EDU}_M^+ = \sum_{M=1}^{W} \mathrm{MAX}(\mathrm{EDU}_M, 0) \\ \mathrm{NEG}(\mathrm{EDU})_{1,it} = \sum_{M=1}^{W} \mathrm{EDU}_M^- = \sum_{M=1}^{W} \mathrm{MIN}(\mathrm{EDU}_M, 0) \end{cases} : \begin{cases} \mathrm{POS}(\mathrm{TO})_{1,it} = \sum_{R=1}^{t} \mathrm{TO}_R^+ = \sum_{R=1}^{T} \mathrm{MAX}(\mathrm{TO}_M, 0) \\ \mathrm{NEG}(\mathrm{TO})_{1,it} = \sum_{R=1}^{t} \mathrm{TO}_R^- = \sum_{R=1}^{T} \mathrm{MIN}(\mathrm{TO}_M, 0) \end{cases} : \begin{cases} \mathrm{POS}(\mathrm{FDI})_{1,it} = \sum_{R=1}^{t} \mathrm{FDI}_R^+ = \sum_{R=1}^{T} \mathrm{MAX}(\mathrm{FDI}_M, 0) \\ \mathrm{NEG}(\mathrm{FDI})_{1,it} = \sum_{R=1}^{t} \mathrm{FDI}_R^- = \sum_{R=1}^{T} \mathrm{MIN}(\mathrm{FDI}_M, 0) \end{cases}$$

Equation (20) distinguishes between long-and short-run asymmetries. The lagged level terms identify whether the cumulative long-run response to positive and negative changes differs, whereas the first-difference terms capture potentially unequal short-run adjustments to the changes. The error-correction coefficient measures the speed at which short-run disequilibrium returns to the long-run relationship.

$$\Delta ED_{it} = \varphi \mathrm{U}_{t-1} + \left(\pi^{+} EDU^{+}_{1,it-1} + \pi^{-} EDU^{-}_{1,it-1}\right) + \left(\beta^{+} TO^{+}_{1,it-1} + \beta^{-} TO^{-}_{1,it-1}\right) + \left(\gamma^{+} FDI^{+}_{1,it-1} + \gamma^{-} FDI^{-}_{1,it-1}\right) + \sum_{j=1}^{m-1} \lambda_j \Delta ED_{it-j_0} + \sum_{r=1}^{n-1} \left(\pi^{+} \Delta EDU^{+}_{1,ir-1} + \pi^{-} \Delta EDU^{-}_{1,ir-1}\right) + \sum_{r=1}^{n-1} \left(\mu^{+} \Delta TO^{+}_{1,ir-1} + \mu^{-} \Delta TO^{-}_{1,ir-1}\right) + + \sum_{r=0}^{m-1} \left(\beta^{+} \Delta FDI^{+}_{1,ir-1} + \beta^{-} \Delta FDI^{-}_{1,ir-1}\right) + \varepsilon_t \quad (20)$$

The error correction term in the above equation is as follows:

$$\Delta \mathrm{IQ}_t = \partial \mathrm{e}_{t-1} + \sum_{j=1}^{m-1} \lambda_j \Delta ED_{it-j_0} + \sum_{r=1}^{n-1} \left(\pi^{+} \Delta EDU^{+}_{1,ir-1} + \pi^{-} \Delta EDU^{-}_{1,ir-1}\right) + \sum_{r=1}^{n-1} \left(\mu^{+} \Delta TO^{+}_{1,ir-1} + \mu^{-} \Delta TO^{-}_{1,ir-1}\right) + + \sum_{r=0}^{m-1} \left(\beta^{+} \Delta FDI^{+}_{1,ir-1} + \beta^{-} \Delta FDI^{-}_{1,ir-1}\right) + \varepsilon_t \quad (21)$$

For cross-sectional unit i and period t, the dynamic SUR model for equation j can be expressed as

$$y_{ijt} = \alpha_j + \sum_{k=1}^{p} \beta_{jk} \gamma_{ijt-k} + \sum_{m=1}^{p} \beta_{jm} \gamma_{ijt-m} + \varepsilon_{ijt} \quad (22)$$

Results

Estimation and interpretation

The findings from the cross-sectional dependency test (Table 3) indicate significant cross-sectional dependence among the variables in the dataset. This implies that the observations are not independent and may exhibit correlations or interdependence between them. The test results, including the LMBP, LMPS, LMadj, and CDPS statistics, indicate statistically significant values, suggesting heteroscedasticity and cross-sectional dependence. These findings underscore the importance of considering the interrelationships between variables when analysing data. Moreover, the test statistics for Δ and Adj.Δ are statistically significant at the 1% level, suggesting heterogeneity in the research variables.

Panel unit root tests were conducted on the following variables: CO_2, C-CO_2, education (EDU), foreign direct investment (FDI), trade openness (TO), and urbanisation (UR). Two specifications were used: one with a constant term and the other with both a constant and trend. CADF and CIPS test statistics were employed. The results in Table 4 indicate that when the level specification includes a constant term, the variables exhibit unit roots, but the values are not statistically significant. However, when we examine the first-difference specification with a constant term, all variables exhibit statistically significant differences.

The study employed a panel cointegration test to document the long-run association between ES, EDU, TO, FDI, and UR by applying cointegration techniques familiarised by Westerlund (2007), Pedroni (2004), and Kao (1999). Table 5 presents the test statistics for the different cointegration tests and shows that all tests reject the null hypothesis of no cointegration. Thus, it is confirmed that there is a long-run association between the variables.

Table 3. Results of cross-sectional dependency and slope heterogeneity.

Variables	LM_{BP}	LM_{PS}	LMadj	CD_{PS}	Δ	Adj.Δ
CO_2	252.362***	15.875***	215.838***	29.852***	32.321***	66.813***
C-CO_2	221.023***	43.447***	157.805***	47.103***	60.76***	82.308***
EDU	280.013***	21.385***	155.827***	27.382***	95.582***	117.524***
FDI	430.819***	17.044***	175.099***	36.406***	27.743***	144.803***
TO	406.557***	29.477***	208.433***	27.074***	74.023***	93.174***
UR	194.251***	21.716***	190.99***	44.642***	20.017***	58.25***

Note: LMBP = Breusch–Pagan LM; LMPS = Pesaran scaled LM; LMadj = bias-adjusted LM; CDPS = Pesaran CD; Δ and Adj. Δ = slope-homogeneity tests. The superscripts of *** denote the level of significance at a 1% level.
***p < 0.01.

Table 4. Results of the second-generation panel unit root test.

Variables	CADF test statistic for constant		CIPS test statistic for constant		CADF test statistic for constant & and trend		CIPS test statistic for constant & and trend	
	Level	1st diff.	Level	1st diff.	Level	1st diff.	Level	1st diff.
CO_2	−1.884	−2.163***	−2.91	−6.75***	−2.476	−6.744***	−2.515	−3.617***
C-CO_2	−1.172	−4.7***	−1.148	−7.439***	−1.477	−7.261***	−1.625	−5.76***
EDU	−2.059	−5.856***	−1.542	−5.111***	−2.466	−6.961***	−1.626	−2.136***
FDI	−2.985	−6.214***	−2.894	−3.288***	−2.775	−6.411***	−2.708	−5.185***
TO	−1.053	−4.086***	−1.498	−3.755***	−1.1	−5.096***	−1.457	−3.517***
UR	−2.412	−4.054***	−2.873	−3.707***	−1.956	−4.828***	−1.453	−5.978***

Note: CADF = cross-sectionally augmented Dickey–Fuller; CIPS = cross-sectionally augmented IPS. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The superscripts of *** denote the level of significance at a 1% level.

Table 5. Results of panel cointegration tests.

Model	EDU →ES	TO →ES	FDI →ES	UR →ES
Gt	−10.7***	−14.465***	−5.546***	−11.74***
Ga	−11.541***	−10.465***	−9.453***	−11.757***
Pt	−12.777***	−10.054***	−9.359***	−10.583***
Pa	−14.404***	−12.623***	−13.992***	−14.028***
KRCPT				
MDF	15.957***	16.157***	3.108***	21.699***
DF	−10.998***	−4.448***	−9.654***	−9.461***
ADF	3.332***	21.439***	−7.126***	5.62***
UMDF	−10.889***	−10.11***	−7.509***	7.549***
UDF	−3.233***	−10.913***	21.301***	−1.126***
PCT				
MDF	12.783***	2.668***	−3.559***	1.162***
PP	8.094***	2.016***	5.959***	8.814***
ADF	8.136***	14.713***	11.538***	10.209***

Note: The superscripts of *** denote significance at the 1% level. The table reports Westerlund, Pedroni, and Kao panel cointegration statistics.
***$p < 0.01$.

In the DSUR model (Table 6), the coefficients for EDU, FDI, TO, and UR are −0.10374, −0.12757, 0.08470, and 0.12987, respectively. The negative signs for EDU and FDI indicate that increases in these variables are associated with reductions in production-based CO_2 emissions, whereas the positive signs for TO and UR indicate emission-increasing associations. The interpretation now follows the signs shown in Table 6 and does not describe a positive coefficient for the emissions reduction.

The CUP-FM estimates preserve the DSUR sign pattern: EDU (−0.14077) and FDI (−0.15352) are associated with lower production-based emissions, whereas trade openness (0.10451) and urbanisation (0.12709) are associated with higher emissions. The CUP-BC estimates are likewise negative for EDU (−0.10075) and FDI (−0.07751) and positive for TO (0.09972) and UR (0.12941). TO denotes trade openness and not technological output, and UR denotes urbanisation and not unemployment. Across the three estimators, the signs were stable, although the magnitudes differed.

Under the DSUR, CUP-FM, and CUP-BC specifications, Table 7 shows that the education expenditure proxy is negatively associated with consumption-based CO_2 emissions, whereas FDI, trade openness, and urbanisation are positively associated. In the DSUR estimates, the coefficients are −0.15816 for EDU, 0.11508 for FDI, 0.13039 for

Table 6. Coefficients of education, trade openness, FDI, and urbanisation on production-based CO_2 emissions.

	Coeff.	Std. Error	t-Statistic	Coeff.	Std. Error	t-Statistic	Coeff.	Std. Error	t-Statistic
	DSUR			CUP-FM			CUP-BC		
EDU	−0.10374	0.0221	−4.6941	−0.14077	0.03	−4.69233	−0.10075	0.0406	−2.48153
FDI	−0.12757	0.0235	−5.4285	−0.15352	0.0392	−3.91633	−0.07751	0.021	−3.69095
TO	0.0847	0.0172	4.9244	0.10451	0.0191	5.471728	0.09972	0.0146	6.830137
UR	0.12987	0.0159	8.1679	0.12709	0.0444	2.862387	0.12941	0.0455	2.844176
C	10.836	0.24013	45.1255	17.565	0.24013	73.14788	16.611	0.24013	69.17503
R^2	0.8904			0.9058			0.8994		
Adj R^2	0.9251			0.9365			0.9263		

Note: DSUR = dynamic seemingly unrelated regression; CUP-FM = continuously updated fully modified; CUP-BC = continuously updated bias-corrected. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 7. Coefficients of education, trade openness, FDI, and urbanisation on consumption-based CO_2 emissions.

	Coeff.	Std. Error	t-Statistic	Coeff.	Std. Error	t-Statistic	Coeff.	Std. Error	t-Statistic
	DSUR			CUP-FM			CUP-BC		
EDU	−0.15816	0.0239	−6.6175	−0.09699	0.0415	−2.33711	−0.09002	0.0376	−2.39415
FDI	0.11508	0.0444	2.5918	0.14942	0.0235	6.3583	0.1194	0.0241	4.95436
TO	0.13039	0.0219	5.9538	0.16387	0.0274	5.980657	0.17669	0.0184	9.602717
UR	0.16056	0.0301	5.3342	0.07819	0.0163	4.796933	0.08383	0.0212	3.954245
C	14.771	0.24013	61.5125	15.34	0.24013	63.88206	17.659	0.24013	73.53933
R^2	0.9105			0.9022			0.8987		
Adj R^2	0.9275			0.9319			0.9463		

Note: C-CO_2 denotes consumption-based carbon dioxide emissions. Estimator and significance notation follow Table 6.

TO, and 0.16056 for UR. Thus, the FDI result for consumption-based emissions is emission-increasing and differs from the negative FDI coefficient for production-based emissions in Table 6.

The CUP-FM and CUP-BC estimates in Table 7 confirm this sign pattern. EDU remains negative (−0.09699 and −0.09002), while FDI (0.14942 and 0.11940), TO (0.16387 and 0.17669), and UR (0.07819 and 0.08383) are positive values. The consistency across estimators supports the directional interpretation, while the magnitude differences caution against presenting a single coefficient as universal across specifications.

Table 8 presents the separate long-run and short-run asymmetric estimates. In the long run, a positive EDU shock is negatively associated with production-based (−0.1079; t = −2.229) and consumption-based emissions (−0.0635; t = −1.960), whereas the corresponding negative shock coefficients are not individually significant at conventional levels. For trade openness and FDI, a positive coefficient on the negative partial-sum variable indicates that a decline in the underlying regressor reduces emissions because the cumulative negative change is multiplied by a positive coefficient, and vice versa. Therefore, the signs must be interpreted jointly with the direction of the shock, rather than treating every positive coefficient as an environmental improvement.

Wald symmetry tests compare the positive and negative shock coefficients. In the long run, symmetry is rejected for EDU, TO, and FDI in the production-based model (W = 8.330, 11.187, and 12.573, respectively). For consumption-based emissions, symmetry is rejected for TO and FDI (W = 4.492 and 10.985) but not for EDU at the 5% level (W = 3.581). In the short run, symmetry is rejected for all three variables in the production-based model (W = 13.570, 8.933, and 4.977, respectively). In the consumption-based model, it is rejected for EDU and FDI (W = 9.143 and 11.106) but not for TO (W = 2.726). These tests show that asymmetry is variable-, horizon-, and emission-measure-specific.

Table 8. Results of the asymmetric assessment.

	CO_2 per capita			Consumption based-CO_2		
Variables	**Coefficient**	**std. error**	**t-stat**	**Coefficient**	**std. error**	**t-stat**
Panel –A: long-run asymmetric coefficients						
EDU^+	−0.1079	0.0484	−2.22934	−0.0635	0.0324	−1.95988
EDU^-	−0.0965	0.0831	−1.16125	−0.0861	0.0814	−1.05774
TO^+	0.0925	0.0347	2.66571	0.0815	0.0769	1.059818
TO^-	0.1263	0.0354	3.5678	0.0765	0.0844	0.906398
FDI^+	0.069	0.0632	1.09177	0.0781	0.0829	0.942099
FDI^-	0.138	0.0759	1.81818	0.0798	0.0816	0.977941
UR	0.123	0.0581	2.11704	0.0667	0.064	1.042188
GCF	0.115	0.043	2.67442	0.1022	0.0507	2.015779
C	0.1076	0.0861	1.24971	0.0737	0.0639	1.153365
Long-run symmetry test						
W_{EDU}	8.33			3.581		
W_{TO}	11.187			4.492		
W_{FDI}	12.573			10.985		
Panel B: short-run asymmetric coefficients						
EDU^+	−0.001	0.00737	−0.13569	−0.0121	0.00415	−2.91566
EDU^-	−0.0359	0.00331	−10.8459	−0.01	0.00376	−2.65957
TO^+	0.0501	0.00434	11.5438	0.0396	0.00431	9.187935
TO^-	0.0346	0.0047	7.3617	0.0283	0.00558	5.071685
FDI^+	−0.0083	0.00778	−1.06684	0.0407	0.00382	10.65445
FDI^-	0.0091	0.00441	2.06349	0.0092	0.00783	1.174968
UR	0.0315	0.00662	4.75831	−0.0001	0.00291	−0.03436
GCF	0.009	0.0035	2.57143	0.0314	0.00359	8.746518
cointEq (−1)	−0.4254	0.00392	−108.52	−0.4367	0.017	−25.6882
Short-run symmetry test						
W_{EDU}	13.57			9.143		
W_{TO}	8.933			2.726		
W_{FDI}	4.977			11.106		

Note: Positive and negative superscripts denote decomposed shocks. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

For production-based emissions, the long-run TO+ and TO− coefficients are positive (0.0925 and 0.1263, respectively), so trade expansion raises emissions, whereas cumulative trade contraction lowers them. The corresponding consumption-based coefficients are also positive (0.0815 and 0.0765), although their individual t-statistics are insignificant. In the short run, the positive and negative TO components are positive and statistically significant for both emissions measures. This evidence indicates that the direction, horizon, and emissions accounting basis jointly determine the environmental interpretation.

The long-run FDI+ and FDI− coefficients are positive for both emissions measures. Accordingly, cumulative increases in FDI are associated with higher emissions, whereas cumulative decreases in FDI are associated with lower emission. However, most individual long-run FDI coefficients are insignificant at conventional levels. Short-run responses differ: FDI+ is not significant for production-based emissions but is positive and significant for consumption-based emissions, whereas FDI− is significant only in the production-based specification. These differences illustrate why short-run asymmetry should not be inferred from long-run coefficients.

Table 9. Dumitrescu–Hurlin panel causality results.

.	CO_2	EDU	TO	FDI	UR	causality
CO_2		(4.458) **	(2.1902) *	(5.3804) ***	(3.5908) **	EDU → CO2; CO2 → UR; FDI → TO; CO2 ←→ TO; CO2 ←→ FDI; EDU ←→ TO; EDU ←→ FDI; EDU ←→ UR; TO ←→ UR; FDI ←→ UR
		[4.6987]	[2.3084]	[5.6709]	[3.7847]	
EDU	1.1923		(4.2656) **	1.5292	(5.5185) ***	
	[1.2567]		[4.496]	[1.6118]	[5.8166]	
TO	(5.7375) ***	1.7747		(3.4367) **	(6.2507) ***	
	[6.0473]	[1.8705]		[3.6223]	[6.5883]	
FDI	(4.6216) **	(5.9245) ***	1.7715		(5.136) ***	
	[4.8712]	[6.2444]	[1.8671]		[5.4133]	
UR	(5.7853) ***	(2.6206) *	(6.0138) ***	(3.8841) **		
	[6.0977]	[2.7621]	[6.3385]	[4.0939]		

Note: The table reports Dumitrescu–Hurlin panel-causality statistics. Arrows in the final column state the direction of the statistically supported relationship. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Based on the Granger causality test results in Table 9, a network of causal relationships emerges among the variables. Education (EDU) is found to Granger-cause carbon emissions (CO_2), indicating a unidirectional causal influence from education to carbon emissions. Likewise, carbon emissions Granger-cause urbanisation (UR), suggesting a one-way causal linkage from CO_2 emissions to UR. Foreign direct investment (FDI) Granger-causes trade openness (TO), indicating a unidirectional causal impact from FDI to TO. Additionally, there are bidirectional causal relationships between CO_2 and trade openness (TO), CO_2 and FDI, EDU and TO, EDU and FDI, EDU and UR, TO and UR, and FDI and UR, indicating mutual causality between these variable pairs. These causal relationships shed light on the complex interplay between education, economic factors, and urbanisation in influencing carbon emissions and environmental sustainability.

Robustness assessment

Table 10 reports the robustness estimates using total greenhouse gas emissions as the dependent variable. The stability analysis supports the principal patterns identified with the carbon dioxide measures: education is negatively associated with emissions, whereas trade openness and urbanisation are positively associated with emissions across the reported estimators. This result is consistent with the interpretation that human capital can facilitate cleaner practices, while scale-related activities remain carbon-intensive in the sample.

FDI is negatively associated with total greenhouse gas emissions across the DSUR, CUP-FM, and CUP-BC specifications (Table 10). This pattern is consistent with a pollution-halo interpretation in the alternative dependent variable test, while the sign differences across other specifications underline the importance of emissions accounting and investment composition.

Table 10. Robustness results using total greenhouse gas emissions.

Variable	DSUR Coeff. (t-Stat)	CUP-FM Coeff. (t-Stat)	CUP-BC Coeff. (t-Stat)	Significance
EDU (Education)	−0.104 (−4.69)	−0.141 (−4.69)	−0.101 (−2.48)	1% level
FDI (Foreign Direct Investment)	−0.128 (−5.43)	−0.154 (−3.92)	−0.078 (−3.69)	1% level
TO (Trade Openness)	0.085 (4.92)	0.105 (5.47)	0.100 (6.83)	1% level
UR (Urbanisation)	0.130 (8.17)	0.127 (2.86)	0.129 (2.84)	1% level
R^2/Adj R^2	0.890/0.925	0.906/0.937	0.899/0.926	—

Note: The dependent variable is total greenhouse gas emissions. Estimator and significance notation follow Table 6.

The contrast between production- and consumption-based CO_2 is substantively significant. Production-based emissions assign carbon to the territory where production occurs, whereas consumption-based emissions reallocate the embodied carbon to the final consumer. The negative FDI coefficients in Table 6 are compatible with the territorial pollution halo channel, but the positive FDI coefficients in Table 7 indicate that domestic territorial improvements may coexist with emissions embodied in imported inputs and consumption. This divergence is consistent with carbon leakage and prevents a simple conclusion on pollution halo. The positive TO coefficients under both accounting measures suggest that scale- and trade-related carbon effects dominate the cleaner technique channel on average in the reported sample.

Table 11 shows stable signs across the alternative estimators. EDU is negative for both production- and consumption-based emissions, with the largest absolute estimate under CCEMG for production-based CO_2 (-0.231). FDI, trade openness, and urbanisation are positive in the CS-ARDL, CCEMG, and AMG specifications. The larger consumption-based FDI estimates are consistent with the embodied emissions associated with import- and trade-linked consumption. These results support the core directional findings while demonstrating that the choice of estimator affects magnitude.

The negative error correction coefficient (approximately -0.43) indicates convergence toward the long-run equilibrium after a short-run disturbance. The CS-ARDL captures dynamic adjustment while augmenting the model for common cross-sectional factors. The CCEMG allows for heterogeneous slopes in the presence of common factors. Finally, the AMG accommodates heterogeneous country coefficients and an unobserved common dynamic process. The agreement in coefficient signs across these estimators indicates that the principal associations are not confined to one estimation method, although the remaining observational limitations warrant cautious interpretation.

Endogeneity assessment using panel System-GMM as a robustness check

Table 12 reports the results of the system-GMM robustness check. DSUR, CUP-FM, CUP-BC, CS-ARDL, CCEMG, and AMG address cross-sectional dependence, common factors, dynamic adjustment, or slope heterogeneity, but they do not eliminate reverse causality by themselves. The System-GMM complements these estimators by including the lagged dependent variable and using internal lag instruments for potentially endogenous regressors. The significant lagged emissions coefficients confirm persistence. The AR(1) results are expected in first-differenced errors, while the non-significant AR(2) tests and Hansen p-values indicate that the reported moment conditions and over-identifying restrictions are not rejected. The instrument counts (29 and 31) remain below the number of cross-sectional units reported for the panel. These diagnostics improve confidence in the robustness of the associations, although they do not convert observational designs into definitive causal evidence.

Table 11. Robustness assessment using alternative estimators.

Variable	CO_2 Emissions (Production-based)			C-CO_2 Emissions (Consumption-based)		
	CS-ARDL	CCEMG	AMG	CS-ARDL	CCEMG	AMG
Education (EDU)	−0.214***	−0.231***	−0.198***	−0.189***	−0.202***	−0.177***
FDI	+0.118***	+0.125***	+0.104***	+0.133***	+0.142***	+0.117***
Trade Openness (TO)	+0.162***	+0.173***	+0.151***	+0.175***	+0.187***	+0.169***
Urbanisation (UR)	+0.091**	+0.085**	+0.078**	+0.094**	+0.089**	+0.083**
Error Correction Term (ECT)	−0.423***	−0.439***	−0.447***	−0.411***	−0.428***	−0.436***
Diagnostic Summary	CD-test $p < 0.01$, Stable long-run, No residual cross-dependence	Heterogeneity accepted	Factor-robust residuals	CD-test $p < 0.01$, Stable long-run	Heterogeneity accepted	Common-factor robustness

Note: CS-ARDL = cross-sectionally augmented autoregressive distributed lag; CCEMG = common correlated effects mean group; AMG = augmented mean group. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 12. Robustness check using the panel system-GMM estimation.

Variable	Coefficient (Model 1: EDU → CO_2)	Std. Error	z-Statistic	P-Value	Coefficient (Model 2: FDI → CO_2)	Std. Error	z-Statistic	P-Value
L.CO_2 (Lagged dependent)	0.612	0.042	14.54	0.000	0.587	0.038	15.42	0.000
EDU	−0.083	0.024	−3.45	0.001	—	—	—	—
FDI	—	—	—	—	0.071	0.028	2.53	0.012
TO	0.049	0.019	2.58	0.010	0.052	0.021	2.47	0.013
UR	0.031	0.013	2.38	0.017	0.029	0.012	2.42	0.015
GCF	−0.026	0.011	−2.31	0.021	−0.022	0.010	−2.20	0.028
Constant	0.378	0.125	3.02	0.003	0.416	0.118	3.53	0.000
Arellano–Bond AR(1) p-val	—	—	—	0.011	—	—	—	0.014
Arellano–Bond AR(2) p-val	—	—	—	0.282	—	—	—	0.336
Hansen J-statistic (p-val)	—	—	—	0.417	—	—	—	0.454
No. of Instruments	29				31			
Observations	1 500				1 500			

Note: System-GMM = system generalised method of moments; AR(1) and AR(2) are Arellano–Bond serial-correlation tests; Hansen is the overidentification test. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 12 further shows that the FDI model has a positive coefficient for foreign investment, indicating that some investment inflows are associated with higher emissions after accounting for the dynamics and endogeneity. This contrasts with the alternative GHG specification, which reinforces the central theoretical point that FDI's environmental effect depends on the type of investment, host country conditions, and emissions measure used. Trade openness and urbanisation remain emission-increasing in the System-GMM specifications, supporting green-conditional investment and trade policies.

The reported coefficients are panel-average effects and may conceal substantial country-level differences in the energy mix, institutional quality, sectoral structure, education systems, and environmental regulation. The significant slope heterogeneity tests in Table 3 and the use of CCEMG/AMG partly accommodate heterogeneous responses; however, the manuscript does not report country-specific coefficients. Therefore, the results should be interpreted as average conditional associations rather than effects that apply uniformly to every country.

Discussion

The education expenditure proxy is negatively associated with both production-based and consumption-based CO_2 emissions in the principal estimates. This finding is consistent with the fiscal capacity channel, in which sustained public education spending can support environmental literacy, technology absorption, and the demand for cleaner policies. Because EDU measures expenditure share rather than attainment or realised human capital, the result should be interpreted as an association with public educational commitment, not as direct evidence that an additional year of schooling reduces emissions.

The rising educational levels of people in sub-Saharan Africa contribute to the knowledge of implementing sustainable practices through positive examples (Ozbay and Duyar, 2022). The transfer of information is vital for spreading green practices across industries, such as the agricultural and power generation sectors. In addition, education enhances economic diversification and the de-industrialisation of polluting economic activities. Those with higher knowledge and experience tend to work in low-carbon industries. High-level education plays a crucial role in developing environmentally conscious industries, including the service sector, high-technology industry, and knowledge economy. The level of

education an individual possesses directly determines their ability to promote strong environmental protection policies (Iorember, Jelilov, Usman, Isik, & Celik, 2021). Based on professional levels, better-educated practitioners can argue in favour of environmentally friendly policies and regulations at all levels of government. Advancements in reducing CO_2 emissions have been impressive, and to some extent, this achievement can be explained by the effective lobbying that led to the implementation of stringent environmental rules, stimulation of environmentally friendly energy sources, and establishment of ambitious emission-cutting objectives (Eyuboglu & Uzar, 2021; Piao & Managi, 2023).

The primary purpose of environmental education programs is to educate students about current environmental issues and encourage them to lead sustainable lives. Learners of formal education programs often engage in environment-protecting activities, such as volunteerism, environmental awareness, and conservation (Thor & Karlsudd, 2020; Boca & Saracli, 2019; Jadhav, Jadhav, and Raut, 2014). Environmental sustainability education raises awareness of the need to conserve natural resources and influences potential action. By incorporating the idea of economic, social, political, and ecological interdependence of urban and rural environments, education can enhance knowledge and motivation, thereby fostering the competence, ideas, and skills required to preserve and improve natural environments (Liu, Gong, and Chen, 2018). Both formal and informal educational programs can alter the attitudes and practices of individuals, communities, and nations toward the environment (Saraclı, Yılmaz, and Arslan 2014). Education for global competency may shape future generations into concerned and active participants in solving global social, political, economic, and environmental issues. Because of its importance in achieving sustainability goals, education is emphasized throughout the 2030 Agenda for Sustainable Development, which urges all nations to "ensure, by 2030, that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of diversity and inclusion. Policies that support and promote sustainability education are crucial for integrating climate change and sustainable development into Indonesia's educational system. These policies may foster a shared understanding of sustainability, establish guidelines for integrating sustainability concepts into educational programs, and support educators' professional development (Abdullah, Zakaria, & Razman, 2018).

Trade openness is positively associated with production- and consumption-based CO_2 emissions across the DSUR, CUP-FM, and CUP-BC estimates in Tables 6 and 7. This pattern supports the scale-effect interpretation: expanded production, transportation, and trade-linked consumption increase emissions. Nevertheless, the asymmetric results show that the response differs between expansions and contractions, and between the short and long run. Cleaner technology diffusion remains a plausible mechanism, but it does not dominate the average coefficient sign reported here.

This study shows that the damages of carbon dioxide emissions can be overcome by increasing openness to trade in various ways. High trade rates can mitigate carbon dioxide emissions as a higher proportion of actors learn to use less polluting technologies and share and access more sustainable materials. This is possible because an increase in bilateral trade may help increase production efficiency, thus reducing the impact on the environment (Rahman, Saidi, and Mbarek, 2020). Weaker trade barriers are associated with lower CO_2 emissions, mediated by a greater level of employment, moderated financial flows, and a more competitive business environment. The rationale for its functioning is the open market system that motivates companies to use cleaner technology. If the adoption of such technology results in an increase in efficiency and output, the economic cost of the environment is likely to decrease (Cherniwchan, Copeland, and Taylor, 2017; Sun et al., 2019). The reduction in carbon dioxide emission has been correlated with trade liberalisation in high- and upper-middle-income economies. Individual nations that have developed technology and environmental management are able to reduce the harmful implications of their economies on the natural environment, hence acquiring the world at the technological level. These standards and technologies can be diffused with the help of trade liberalisation, and other countries can reduce their environmental effects (Farhani, Chaibi, & Rault, 2014; Pao and Tsai, 2010).

The environmental association of FDI depends on emissions measures and specifications. In the principal symmetric estimates, FDI is negatively associated with production-based emissions and positively associated with consumption-based emissions. Alternative estimators and System-GMM yield positive FDI coefficients, whereas the GHG robustness specification yields negative coefficients. This variation is consistent with competing pollution havens, pollution halos, scales, and embodied carbon channels. This also indicates that the study should not characterise FDI as uniformly beneficial or harmful; the net association depends on investment composition, host-country regulation, technology transfer, and where emissions are accounted for.

Conclusions and policy implications

Conclusion

This study examines the dynamic relationships among government education expenditure share, trade openness, FDI, and environmental sustainability in the reported Sub-Saharan African panel. The education expenditure proxy is

negatively associated with both emissions measures; however, this result should not be interpreted as a direct attainment effect. Trade openness and urbanisation generally increase emissions. FDI produces mixed evidence: it is negatively associated with territorial production-based emissions in the principal long-run estimates but is positively associated with consumption-based emissions and in several alternative specifications. The asymmetric estimates further show that increases and decreases can generate different short- and long-run responses. These findings support policies that combine sustained educational investment with environmental screening of FDI, cleaner trade logistics, embodied carbon monitoring, and country-specific regulatory design.

Policy suggestion

In light of the environmental issues that Sub-Saharan Africa (SSA) faces and to harness the opportunities in education, trade openness, and foreign direct investment (FDI) to improve environmental sustainability, the following policy formulations are proposed: First, educational programs specifically aimed at fostering environmental awareness and adopting environmentally sustainable practices should be strengthened. The appropriate approach is to make sustainability a central idea in the mainstream curriculum at all levels, from primary to tertiary education. Policymakers should introduce better teacher training and create instructional materials that outline the importance of individual environmental custodianship. Moreover, allocating resources to adult training and education campaigns at the community level may significantly increase their outreach and efficacy. International development agencies must work closely with local budgetary mechanisms to secure the funds required to implement educational reforms. The expected result is an informed populace able to advance sustainable measures at both personal and organizational scales, thereby driving a decrease in CO_2 levels and enhancing environmental standards for all.

Second, advertising and green investments should be increased to counter the environmental impacts of FDI and trade openness, which are the main determinants of economic growth. Governments should consider implementing FDI laws that compel foreign investors to adhere to environmental performance standards. This supports the notion that environmental considerations are systematically considered when making investment decisions. The only solution would be to require the integration of green technologies and environmentally friendly mechanisms as part of the investment permits. These regulatory changes should be implemented through the consolidation of enforcement systems and the involvement of citizens, businesses, and non-bank financial institutions. In addition, trade in environmentally friendly goods and services through interventions should be promoted to offset the environmental externalities arising from trade liberalisation on global CO_2 emissions. Through these steps, policymakers will be able to improve economic growth, minimise environmental degradation, and develop a more balanced developmental paradigm for the future. These advantages are expected to result in better environmental standards, sustainable economic development, and an increase in social welfare, which makes these reforms appealing to the political establishment as well as to society.

This study had several limitations. First, EDU is a fiscal-input measure that does not directly capture attainment, literacy, schooling years, education quality, environmental knowledge, or realised human capital. Second, although UR and GCF capture the urban scale and domestic investment channels, omitted factors such as energy mix, industrial composition, regulatory enforcement, institutional quality, and environmental policy may still affect the estimates. Third, DSUR, CUP-FM, CUP-BC, CS-ARDL, CCEMG, AMG, and System-GMM address different forms of dependence, dynamics, heterogeneity, and endogeneity; however, the observational design does not establish definitive causality. Fourth, the panel-average coefficients may conceal country-specific effects despite the heterogeneous panel estimators. Future work should test alternative education measures, additional lag structures and control sets, country or income group heterogeneity, and sectoral or bilateral embodied carbon channels. Therefore, the regional and period-specific results should be generalised with caution.

Ethical considerations

Not applicable. This study used publicly available aggregate secondary data and did not involve human participants, animals, interviews, surveys, or identifiable data.

Data availability

This study uses publicly accessible data from the World Development Indicators (World Bank), UNCTAD, and Our World in Data (OWID). The processed dataset and associated materials are available at: qamruzzaman, Md (2026). Reassessing the Education–FDI–Trade Nexus for a Sustainable Future: Dataset. Figshare. Dataset. https://doi.org/10.6084/m9.figshare.33070697. Data are available under the terms of the Creative Commons Attribution 4.0 International license (CC-BY 4.0).

Reporting guidelines

No EQUATOR reporting guideline is specifically applicable because this is an econometric study based solely on aggregate publicly available secondary panel data. This manuscript reports the data sources, variable definitions, sample period, and estimation procedures required for replication.

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Reviewer Report 27 August 2026

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The manuscript addresses an important and timely issue. The topic is relevant and potentially publishable; however, the manuscript requires substantial revision before it can be considered suitable for indexing.

- There are numerous grammatical, typographical, and terminology problems that reduce readability.
- The manuscript states that it investigates Sub-Saharan African economies, while the methods section refers to the “top 50 polluted nations”. The authors should clearly identify the countries included in the final balanced/unbalanced panel, explain the selection criteria, and report the number of countries and observations.
- Education is described conceptually as educational attainment/human capital, but Table 2 measures EDU as education spending as a share of total government expenditure. These are not equivalent measures. The authors should justify why government education expenditure is an appropriate proxy for education and distinguish expenditure from actual educational attainment, human capital, literacy, or years of schooling.
- There are important inconsistencies between the reported coefficients and their interpretations. For example, Table 6 reports positive coefficients for EDU and FDI, while the accompanying text interprets these coefficients as reductions in emissions.
- The authors should explain why positive and negative shocks are expected to have different environmental consequences and clearly distinguish long-run asymmetry from short-run asymmetry.
- The symmetry tests should also be interpreted systematically rather than simply reported.
- The authors should explain why the selected estimators adequately address reverse causality and omitted-variable bias. If possible, additional robustness checks using alternative specifications or lag structures should be reported.
- The authors should provide a stronger theoretical and empirical justification for the final control-variable specification. Otherwise, omitted-variable bias may affect the estimated coefficients.

- Using both production-based and consumption-based CO_2 emissions is a strength of the paper. However, the manuscript should more clearly explain what the differences between these two dependent variables imply for the pollution-haven, pollution-halo, and carbon-leakage arguments.
- The authors should discuss whether the estimated average effects may conceal substantial country-level differences.
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Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and is the work technically sound?
Partly

Are sufficient details of methods and analysis provided to allow replication by others?
Partly

If applicable, is the statistical analysis and its interpretation appropriate?

Partly

Are all the source data underlying the results available to ensure full reproducibility?
Partly

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: FDI, digital trade, environmental science

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 28 Aug 2026

Md Qamruzzaman

Dear Editor and Reviewer,
We sincerely thank the reviewer for the constructive and detailed comments. We have revised the manuscript carefully while preserving the original text as much as possible. All changes in the revised manuscript are shown in red ink for convenience. Because the reviewer file repeated the same set of comments, the duplicated comments have been consolidated below and each unique point is answered separately.
Comment 1
Reviewer comment: *There are numerous grammatical, typographical, and terminology problems that reduce readability.*
Response: We thank the reviewer for this observation. The manuscript has been carefully edited for grammar, typographical accuracy, terminology consistency, and readability. We also corrected corrupted wording and inconsistent terminology in the introduction, literature review, methods, empirical-results discussion, conclusion, and limitations sections.
Manuscript change made: Language and terminology were revised throughout the manuscript. The revisions were made conservatively to preserve the original text as much as possible, and all changes were marked in red ink in the revised manuscript.
Comment 2
Reviewer comment: *The manuscript states that it investigates Sub-Saharan African economies, while the methods section refers to the 'top 50 polluted nations'. The authors should clearly identify the countries included in the final balanced/unbalanced panel, explain the selection criteria, and report the number of countries and observations.*
Response: We agree that the earlier wording was inconsistent and could mislead readers about the geographical scope of the study. The manuscript now states clearly that the study concerns Sub-Saharan African economies and that the reported estimation sample is a balanced panel of 50 cross-sectional country units over 1990-2019, yielding 1,500 country-year observations.
Manuscript change made: The inconsistent phrase 'top 50 polluted nations' was removed.

The data section now reports the 50-country balanced panel, the 1990-2019 period, the 1,500 observations, and the selection criterion based on the availability of the variables required for the final specifications. An author-verification note remains in red requesting insertion of the exact 50 country names and ISO codes from the deposited replication dataset before resubmission.

Comment 3

Reviewer comment: *Education is described conceptually as educational attainment/human capital, but Table 2 measures EDU as education spending as a share of total government expenditure. These are not equivalent measures. The authors should justify why government education expenditure is an appropriate proxy for education and distinguish expenditure from actual educational attainment, human capital, literacy, or years of schooling.*

Response: We appreciate this important clarification. The revised manuscript no longer treats government education expenditure as equivalent to educational attainment or realized human capital. Instead, EDU is explicitly presented as a fiscal-input proxy that reflects public commitment to education and the capacity to finance schools, teachers, curricula, and environmental learning.

Manuscript change made: Table 2 and the model discussion now define EDU as government education expenditure as a share of total government expenditure. The text explicitly states that EDU is not a direct measure of literacy, years of schooling, education quality, attainment, or realized human capital. The discussion and conclusion were revised accordingly.

Comment 4

Reviewer comment: *There are important inconsistencies between the reported coefficients and their interpretations. For example, Table 6 reports positive coefficients for EDU and FDI, while the accompanying text interprets these coefficients as reductions in emissions.*

Response: We agree that the coefficient interpretations needed to be aligned precisely with the reported signs. The empirical-results section has been revised so that each interpretation follows the coefficient signs shown in the corresponding tables and no positive coefficient is described as an emissions reduction.

Manuscript change made: The discussion of Tables 6 and 7 was rewritten. The revised text states that EDU and FDI are negatively associated with production-based CO2 emissions in the principal estimates, while trade openness and urbanisation are positively associated with emissions. For consumption-based CO2 emissions, EDU remains negative, while FDI, trade openness, and urbanisation are positive. The abstract and conclusion were also adjusted to reflect these corrected interpretations.

Comment 5

Reviewer comment: *The authors should explain why positive and negative shocks are expected to have different environmental consequences and clearly distinguish long-run asymmetry from short-run asymmetry.*

Response: We thank the reviewer for this helpful suggestion. The revised manuscript now provides a clearer theoretical explanation of asymmetry. It explains that increases and decreases in education expenditure, trade openness, and FDI may have different environmental effects because of adjustment costs, policy responses, sectoral composition, capital-stock adjustment, and technology-transfer channels.

Manuscript change made: The asymmetric-model section now explains the decomposition into cumulative positive and negative partial sums. The text distinguishes long-run asymmetry, captured by lagged level terms, from short-run asymmetry, captured by first-

difference terms. The interpretation of Table 8 was revised to separate shock direction, time horizon, and emissions-accounting basis.

Comment 6

Reviewer comment: *The symmetry tests should also be interpreted systematically rather than simply reported.*

Response: We agree. The revised manuscript now interprets the Wald symmetry tests in a systematic way rather than only listing the statistics. The interpretation identifies which variables show long-run and short-run asymmetry and explains that asymmetry is variable-specific, horizon-specific, and emissions-measure-specific.

Manuscript change made: A new explanatory paragraph was added after Table 8. It interprets the Wald tests for EDU, trade openness, and FDI separately for production-based and consumption-based CO2 emissions in both the long run and short run.

Comment 7

Reviewer comment: *The authors should explain why the selected estimators adequately address reverse causality and omitted-variable bias. If possible, additional robustness checks using alternative specifications or lag structures should be reported.*

Response: We appreciate the reviewer's concern. The revised manuscript now explains the role and limits of each estimator more carefully. It clarifies that DSUR, CUP-FM, CUP-BC, CS-ARDL, CCEMG, and AMG address cross-sectional dependence, common factors, dynamic adjustment, and slope heterogeneity, but they do not alone eliminate reverse causality. System-GMM is presented as a complementary robustness check using lagged dependent variables and internal lag instruments.

Manuscript change made: The robustness and endogeneity-assessment sections were revised. The System-GMM discussion now reports the interpretation of the lagged dependent variable, AR(1), AR(2), Hansen tests, and instrument counts. The wording was also adjusted to avoid overstating causal identification from an observational panel design.

Comment 8

Reviewer comment: *The authors should provide a stronger theoretical and empirical justification for the final control-variable specification. Otherwise, omitted-variable bias may affect the estimated coefficients.*

Response: We agree that the control specification needed clearer justification. The revised manuscript now explains why urbanisation and gross capital formation are included as controls. Urbanisation captures urban scale, density, transport, construction, and energy-demand channels, while gross capital formation captures domestic investment-driven expansion that may affect emissions independently of FDI.

Manuscript change made: The methods section now identifies EDU, FDI, and trade openness as focal regressors and urbanisation and gross capital formation as theoretically motivated controls. The text also acknowledges that these controls reduce but cannot eliminate omitted-variable bias because unobserved factors such as energy mix, industrial structure, institutional quality, regulation, and environmental policy may still matter.

Comment 9

Reviewer comment: *Using both production-based and consumption-based CO2 emissions is a strength of the paper. However, the manuscript should more clearly explain what the differences between these two dependent variables imply for the pollution-haven, pollution-halo, and carbon-leakage arguments.*

Response: We thank the reviewer for recognizing this strength. The revised manuscript now explains that production-based emissions assign carbon to the territory where

production occurs, while consumption-based emissions reallocate embodied carbon to final consumption. This distinction is important because territorial improvements may coexist with imported or consumption-embodied emissions.
Manuscript change made: A new discussion was added explaining how the different results for production-based and consumption-based CO2 emissions affect the interpretation of pollution-haven, pollution-halo, and carbon-leakage mechanisms. The revised text avoids a simple pollution-halo conclusion and explains why FDI and trade effects depend on where emissions are accounted for.
Comment 10
Reviewer comment: *The authors should discuss whether the estimated average effects may conceal substantial country-level differences.*
Response: We agree. The revised manuscript now states that the reported coefficients are panel-average effects and may conceal substantial country-level heterogeneity in energy mix, institutional quality, sectoral structure, education systems, and environmental regulation.
Manuscript change made: A paragraph was added to the empirical discussion explaining the limits of panel-average interpretation. The limitations section was also expanded to recommend future testing of country-level, income-group, sectoral, and embodied-carbon heterogeneity.

Competing Interests: no

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